

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech.

Branch: Mechanical Engg.

Semester: VII

Subject Code & Name: BTMEC702 CAD/CAM

Max Marks: 60

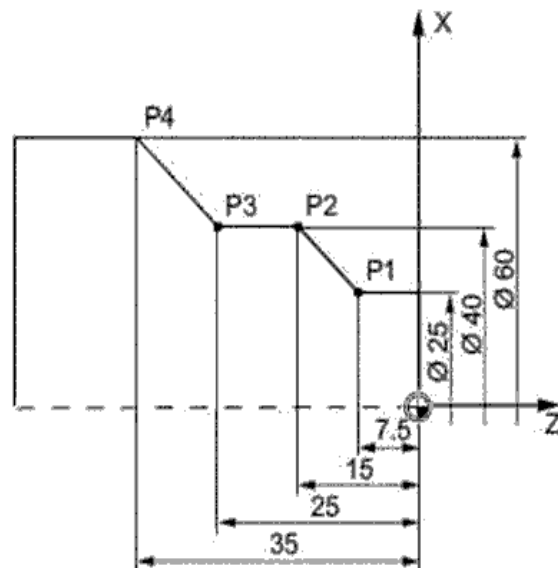
Date: 15/07/2023

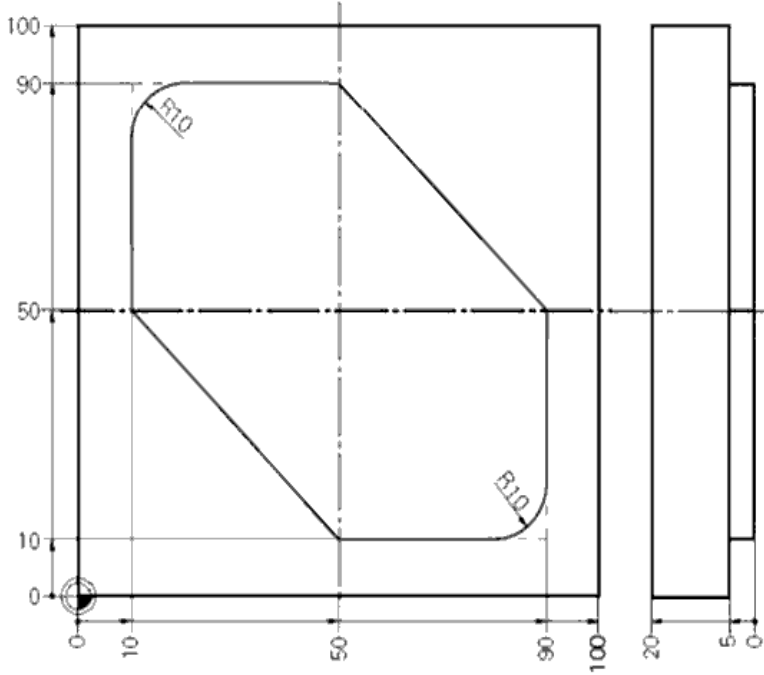
Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain different types of plotters used in CAD applications. Also explain the relative advantages and limitations of each.	CO-01	6
B)	Explain various ground rules for a graphics software?	CO-02	6
C)	Describe any three methods for defining arc of a circle in a graphics software.	CO-01	6
Q.2	Solve Any Two of the following.		12
A)	Compare the wireframe, surface and solid modeling as the modeling techniques for CAD.	CO-03	6
B)	Explain the need of synthetic curves in the CAD applications.	CO-03	6
C)	What is FMS? Explain various components of a typical FMS system. Explain the advantages of FMS in manufacturing advantages.	CO-07	6
Q. 3	Solve Any Two of the following.		12
A)	Differentiate between NC, CNC and DNC systems?	CO-05	6
B)	Write NC program for finishing operation of turning of the job as shown in following figure.	CO-05	6



C)	Write NC program for finishing operation of milling of outer surface of the job as shown in following figure.  All the dimensions are in mm	CO-05	6
Q.4	Solve Any Two of the following.		12
A)	Explain the steps involved in solving a problem using Finite Element Analysis.	CO-06	6
B)	Explain various properties of stiffness matrix	CO-06	6
C)	Explain the elements used in structural analysis using FEM. (at least one element each from 1D, 2D and 3D)	CO-06	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the advantages of GT layout in manufacturing industry.	CO-07	6
B)	Explain the concept of “composite Part” in GT using a suitable example.	CO-08	6
C)	Compare generative and retrieval type of CAPP systems.	CO-08	6
	*** End ***		